Photonic QRC for Swaption Volatility Forecasting

Temporal reservoir computing on a photonic processor — no gradient descent, no barren plateaus

Fanot Qubits
Quandela Hackathon 2026
Quandela MerLin SDK

0.000636	19×	8	65	224
BEST QLIKE	VS MLP	PHOTONIC MODES	HYBRID FEATURES	VOL POINTS

PROBLEM

Forecast a 14-tenor × 16-maturity swaption implied volatility surface (224 points) from 5-day historical windows. Only **489 training samples** are available, making deep recurrent models unreliable. Primary metric: **QLIKE** — the financially correct volatility loss that penalises under-prediction asymmetrically.

FINAL ARCHITECTURE

Surface 224-dim → PCA → 5 comp. → Photonic QRC 8 modes · 3 photons → LexGrouping Fock → 8 feat. → Ridge α=0.1 analytical

Mode split: 5 input modes receive angle-encoded PCA features per timestep; 3 memory modes accumulate temporal state — photonic realisation of Li et al. (arXiv:2505.13933).

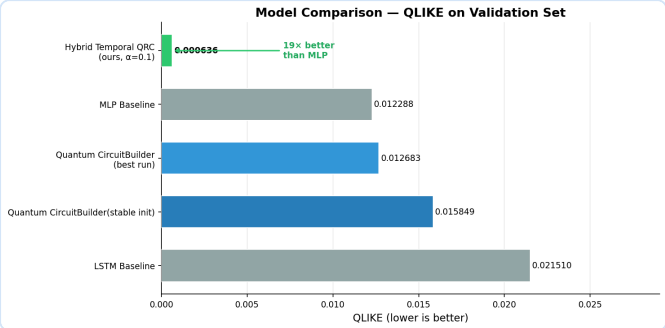
Feature vector: 5 timesteps × 8 Fock measurements = 40 quantum + 25 classical PCA = **65 hybrid features**.

Fixed reservoir: circuit weights frozen after random init — no gradient computation, no barren plateaus, deterministic every run.

BENCHMARK RESULTS

MODEL	RMSE	MAE	QLIKE
🥇 **Hybrid Temporal QRC (ours)**	**0.003467**	**0.002591**	**0.000636**
MLP Baseline	0.011867	0.008495	0.012288
Quantum CircuitBuilder	0.014969	0.009463	0.012683
Quantum (stable init)	0.014574	0.010024	0.015849
LSTM Baseline	0.017637	0.013449	0.021510

MODEL COMPARISON



QLIKE across all five models — lower is better. 19× gap between our result and the MLP baseline.

DEVELOPMENT JOURNEY

STEP 1
Classical Baselines
MLP vs LSTM under identical conditions. LSTM fails with 489 samples. MLP sets ceiling.
MLP 0.012288

STEP 2
First Quantum Model
QuantumLayer.simple() + MSE loss. 16-mode circuit, LexGrouping, MLP readout.
RMSE 0.01959

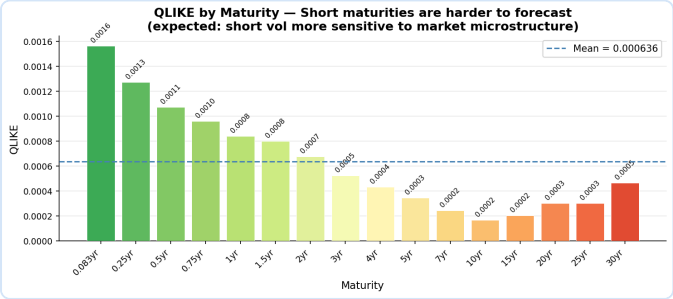
STEP 3
CircuitBuilder + QLIKE
Explicit CircuitBuilder (scale=π) gives 2× boost. QLIKE loss. High variance across seeds.
Best 0.012683

STEP 4
Barren Plateau
±50% variance = lucky seeds. Identity-block init (Grant et al. 2019) stabilises but caps.
Stable 0.015849

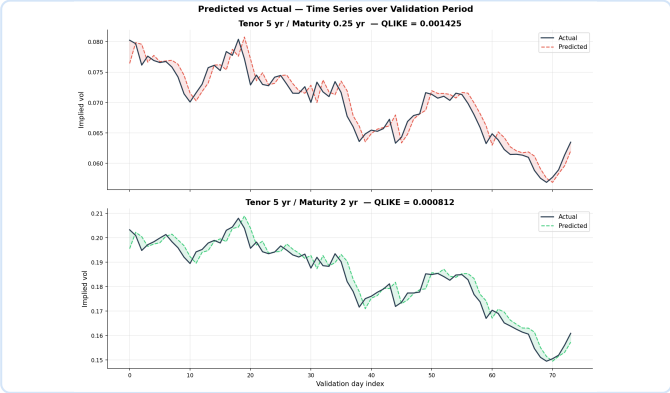
STEP 5
Temporal QRC Pivot
Li et al. fixed reservoir. 5+3 mode split, sequential encoding, Ridge. No gradient needed.
QRC 0.000713

STEP 6
Alpha Tuning
Ablation over α ∈ {0.0001…100}. α=0.1 unlocks quantum feature contribution.
Final 0.000636

VISUALISATIONS



QLIKE by maturity — error distribution across the 16 maturities on the swaption surface.



Predicted vs actual — implied vol time series for three representative surface cells.